LECTURE 02 · STUDY NOTES

# Leveraged Investment Strategies

*Margin accounts, leveraged ETFs, short selling and inverse funds, and why the math of compounding changes everything.*

**How to use these notes**

These notes follow the order of the Lecture 02 slides, but they explain each idea in full sentences rather than slide fragments. Every numerical exercise from class is worked out step by step, with each intermediate number shown so you can check your own work line by line. Try each example on your own first, then compare. Practice problems with full solutions are at the end.

## 1. The opening question: what does leverage do to a real market path?

The lecture opened with the path of the three major U.S. stock indexes from January 2019 through September 2026, each rescaled so that January 2019 equals 100. Over that window the NASDAQ-100 (tracked by the QQQ ETF) gained **+260.5%**, the S&P 500 (SPY) gained **+182.4%**, and the Dow Jones Industrial Average (DIA) gained **+109.5%**. The gap is mostly about composition: the NASDAQ-100 is concentrated in large technology firms, while the Dow carries more industrials, energy and financials.

The same chart also shows the other side of that return. In 2022 the NASDAQ-100 fell by roughly a third. Higher long-run returns came with deeper drawdowns along the way. That sets up the question for the rest of the lecture: if an investor had used borrowed money, or a 3× leveraged ETF, on this path, what would have happened? The answer depends on much more than just multiplying the return by three.

## 2. Recap of Lecture 01: ETFs, mutual funds and NAV

An investment company pools money from many investors and buys a portfolio. Its per-share value is the **net asset value**: what the fund owns, minus what it owes, divided by the number of shares outstanding.

$$\text{NAV} = (\text{Assets} - \text{Liabilities}) \div \text{Shares outstanding}$$

WORKED EXAMPLE 1

### Computing NAV

A fund holds $500 million of stocks, owes $5 million (accrued fees and other liabilities), and has 25 million shares outstanding.

| Step | What we compute | Result |
|---|---|---|
| **1** | Net assets = Assets − Liabilities = $500M − $5M | **$495M** |
| **2** | Divide by shares outstanding: $495M ÷ 25M shares | **NAV = $19.80** |

Mutual funds and ETFs differ in how you trade them. A mutual fund share is bought from or sold back to the fund once a day at that day's closing NAV, so the price you get *is* the NAV. An ETF share trades on an exchange all day at a market price set by buyers and sellers, and the fund is not a party to most of those trades. That price can drift above NAV (a premium) or below it (a discount).

**Authorized participants** (APs), large dealers with a special agreement with the fund, keep that gap small. If the ETF trades above NAV, an AP delivers the underlying basket to the fund, receives new ETF shares worth NAV, and sells them on the exchange at the higher price. That extra supply pushes the price back down. If the ETF trades below NAV, the AP does the reverse: it buys cheap ETF shares, redeems them for the underlying basket worth NAV, and the buying pressure pushes the price up. For liquid funds such as SPY or QQQ the gap is usually 0.01–0.05%; during market stress it can widen to 1–3%.

| | ETF | Mutual fund |
|---|---|---|
| Trading | Intraday on an exchange | Once a day at NAV |
| Price you get | Market price | End-of-day NAV |
| Minimum investment | One share | Often $1,000+ |
| Tax efficiency | High (in-kind redemptions) | Lower (capital-gain distributions) |
| Expense ratio | Very low (0.03–0.20%) | Higher (0.5–1.5%) |
| Can be sold short | Yes | No |

This matters for today because leveraged ETFs use exactly the same ETF wrapper. What they add is debt and derivatives, and a NAV that is reset to the target leverage every single night.

## 3. Part A: Leveraged ETFs, the TQQQ case

The case study is the ProShares UltraPro QQQ fund, ticker **TQQQ**. It launched in February 2010, manages roughly $22 billion (2025), and is the largest leveraged ETF in the world. Its objective is to deliver **three times the daily return** of the NASDAQ-100. It charges an

expense ratio of 0.88%, compared with 0.20% for QQQ, which tracks the same index without leverage.

The word *daily* is the most important word in that description. TQQQ promises 3× the return of each individual trading day, not 3× the return over a week, a year, or your holding period. To keep that promise, the fund rebalances at every market close so that its exposure is again exactly three times its NAV. As Part C shows, that nightly reset makes multi-day results differ, sometimes dramatically, from three times the index's multi-day return.

## How the fund actually creates 3× exposure

TQQQ does not buy three dollars of Apple, Microsoft and NVIDIA for every dollar investors give it. It builds the exposure synthetically. About 65% of the exposure comes from **total return swaps**: over-the-counter contracts with large banks in which the bank pays the fund the NASDAQ-100's return and the fund pays the bank a floating financing rate (SOFR plus a spread). About 25% comes from **E-mini NASDAQ-100 futures**, exchange-traded contracts that are rolled every quarter. The remaining assets sit in **Treasury bills**, which serve as collateral for the swaps and earn short-term interest.

This is why the holdings list of TQQQ looks nothing like QQQ's. QQQ's top holdings are Apple (9.0%), Microsoft (8.2%), NVIDIA (8.0%) and the rest of the index, with its top 20 names making up about 68% of the fund. TQQQ's top holdings are six T-bills making up roughly 85% of net assets, followed by swap contracts with Goldman Sachs, Morgan Stanley, Citibank, BNP Paribas, UBS and others, and a small futures position. It owns *no stocks at all*. The swaps appear at only a few percent of net assets because a swap is carried at its unrealized gain or loss, not at the size of the exposure it creates.

**WORKED EXAMPLE 2**

## Holdings versus exposure

Suppose TQQQ's NAV is $100 per share and the NASDAQ-100 rises 1% today. What happens, and how does the fund get back to 3× for tomorrow?

| Step | What we compute | Result |
|---|---|---|
| **1** | Required exposure at the start of the day = 3 × NAV = 3 × $100 | **$300 of index exposure** |
| **2** | Where it comes from: the fund's own $100 plus about $200 of *notional* exposure financed through swaps and futures | **Swap notional ≈ 200% of NAV** |
| **3** | Index rises 1%: gain on exposure = 1% × $300 | **+$3.00** |
| **4** | New NAV = $100 + $3 (ignoring one day of fees and financing) | **$103.00 (+3.0%)** |
| **5** | Exposure after the move = $300 × 1.01. Target for tomorrow = 3 × $103 | **$303 held vs. $309 needed** |
| **6** | Rebalance at the close: add exposure equal to $309 − $303 | **Buy $6 more exposure** |

Step 6 is the key mechanical point. After an *up* day, a leveraged ETF must *add* exposure; after a *down* day it must *cut* exposure. Leveraged funds are therefore forced buyers into strength and forced sellers into weakness, always near the close. The same logic makes the cost of leverage explicit: the fund pays a financing rate on the borrowed $200 of exposure every day it is held, on top of the 0.88% management fee.

### Why regulators pay attention

Several regulators have flagged these products. The IMF's *Global Financial Stability Report* (October 2023) described leverage in non-bank financial intermediation as a key vulnerability and pointed out that end-of-day rebalancing by leveraged ETFs mechanically amplifies directional moves. The SEC staff (November 2022) warned retail investors that leveraged and inverse ETFs are complex products typically not suitable for long-term holding, and brokers must now assess suitability before recommending them. FINRA (2023) found that brokers often treated these funds like ordinary ETFs and rarely explained the daily compounding effect to clients.

The class discussion grouped the concerns into three channels. First, **forced rebalancing**: as Example 2 shows, the whole leveraged-ETF sector buys after rallies and sells after declines, which adds momentum to the market. Second, **counterparty concentration**: the fund's exposure sits with a small group of large banks, so stress at those banks reaches the fund immediately. Third, the **liquidity illusion**: TQQQ shares trade $3–4 billion a day in normal markets, but the swaps underneath are OTC contracts that could gap in a crash. The liquidity of the ETF share is not the same thing as the liquidity of what it holds.

## 4. Part B: How debt amplifies returns

To see leverage without any ETF machinery, compare two investors who each control a $10,000 position in the same asset. The unleveraged investor puts up all $10,000. The

leveraged investor puts up $5,000 and borrows the other $5,000 at 6% per year, a 2:1 ratio of assets to equity. Both own exactly the same asset; the only difference is who financed it. The lender's claim is **senior**: principal plus interest ($5,000 × 6% = $300) must be repaid whatever the asset does, and the investor keeps only what is left, the **residual**.

WORKED EXAMPLE 3

### Same $10,000 position, bull (+20%) and bear (−20%)

| Step | What we compute | Result |
|---|---|---|
| **1** | Capital structure. Unleveraged: $10,000 equity, $0 debt. Leveraged: $5,000 equity, $5,000 debt at 6% | **Both invest $10,000** |
| **2 · Bull** | Asset value after +20%: $10,000 × 1.20 | **$12,000 (both)** |
| **3 · Bull** | Leveraged repays the lender: $5,000 + 6% × $5,000 | **$5,300** |
| **4 · Bull** | Unleveraged ROE = ($12,000 − $10,000) ÷ $10,000 | **+20%** |
| | Leveraged: keeps $12,000 − $5,300 = $6,700. Profit = $6,700 − $5,000 = $1,700. ROE = $1,700 ÷ $5,000 | **+34%** |
| **2 · Bear** | Asset value after −20%: $10,000 × 0.80 | **$8,000 (both)** |
| **3 · Bear** | Lender is still owed principal plus interest | **$5,300** |
| **4 · Bear** | Unleveraged ROE = ($8,000 − $10,000) ÷ $10,000 | **−20%** |
| | Leveraged: keeps $8,000 − $5,300 = $2,700. Loss = $2,700 − $5,000 = −$2,300. ROE = −$2,300 ÷ $5,000 | **−46%** |

The asset moved the same ±20% for both investors, yet the leveraged investor earned +34% or lost −46%. The effect is also *asymmetric*: the upside was amplified by 14 points and the downside by 26 points, because the $300 of interest is paid in both states. The same result comes from a formula you can use for any leverage ratio:

$$\text{ROE} = r + (D \div E) \times (r - i)$$

where $r$ is the asset's return, $i$ the borrowing rate, and D ÷ E the debt-to-equity ratio (here $5,000 ÷ $5,000 = 1). Bull: 20% + 1 × (20% − 6%) = 34%. Bear: −20% + 1 × (−20% − 6%) = −46%. The term (r − i) tells you when leverage helps: only when the asset earns more than the cost of borrowing. The break-even is r = i = 6%. At exactly 6% both investors earn 6% on their equity; below 6%, borrowing makes the leveraged investor worse off.

# 5. Part B: Buying on margin and the margin call

A margin account is how an individual investor borrows from a broker to buy securities. Two percentages govern it. The **initial margin** is the minimum share of the purchase you must fund with your own money; under the Federal Reserve's Regulation T it is 50% for stocks. The **maintenance margin** is the minimum equity you must keep afterwards; the class example uses 30%, and brokers often set their own higher levels. At any moment, the margin in the account is:

$$\text{Margin} = \text{Equity} \div \text{Market value} = (P \times Q - \text{Loan}) \div (P \times Q)$$

The loan is fixed in dollars, while the value of the shares moves with the price. So every price decline comes entirely out of the investor's equity, and the margin percentage falls faster than the price does.

**WORKED EXAMPLE 4**

## Buying 100 shares at $10 with a $500 loan

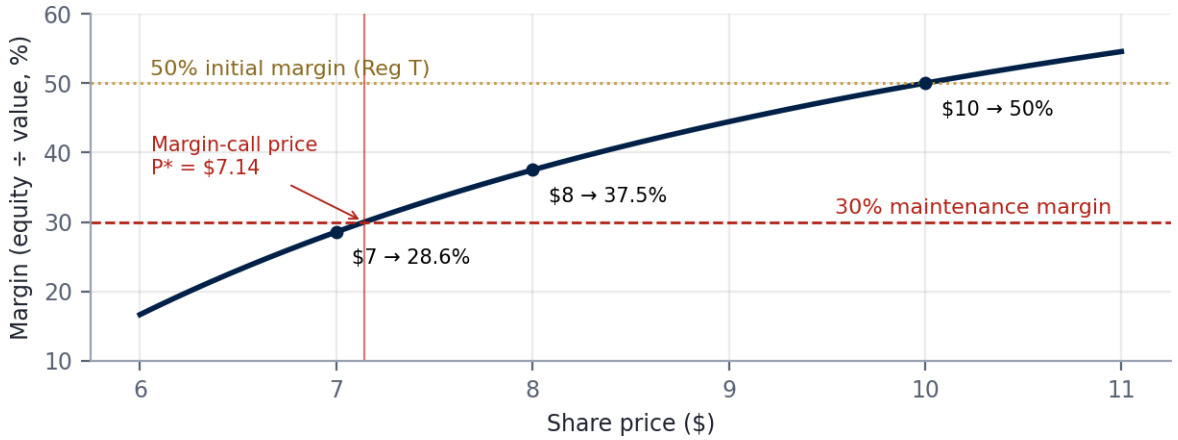
| Step | What we compute | Result |
|---|---|---|
| **1 · Open** | Position value = 100 × $10 | **$1,000** |
| | Equity = $1,000 − $500 loan | **$500** |
| | Margin = $500 ÷ $1,000 (Reg T minimum is 50%) | **50%, allowed** |
| **2 · $8** | Position value = 100 × $8; loan unchanged at $500 | **$800** |
| | Equity = $800 − $500 | **$300** |
| | Margin = $300 ÷ $800 | **37.5%, above 30%** |
| **3 · $7** | Position value = 100 × $7 | **$700** |
| | Equity = $700 − $500 | **$200** |
| | Margin = $200 ÷ $700 | **28.6%: MARGIN CALL** |

Notice what happened between the open and $7. The price fell 30%, but the investor's equity fell from $500 to $200, a 60% loss. That is the 2:1 leverage from Example 3 again. With margin, though, the losses do not just accumulate: once the margin drops below maintenance, the broker issues a **margin call**. The investor must add cash or securities immediately, or the broker may sell the shares without further notice.

## Finding the margin-call price

Instead of testing prices one at a time, we can solve for the exact price P* at which the margin equals the maintenance requirement m. Set the margin formula equal to m and solve for P:

| Step | Algebra | Result |
|---|---|---|
| **1** | Start from the definition and set it equal to m | **(P×Q − L) ÷ (P×Q) = m** |
| **2** | Multiply both sides by P×Q | **P×Q − L = m × P×Q** |
| **3** | Collect the P terms on one side | **P×Q × (1 − m) = L** |
| **4** | Divide by Q × (1 − m) | **P* = L ÷ [Q × (1 − m)]** |
| **5** | Plug in L = \$500, Q = 100, m = 0.30: \$500 ÷ (100 × 0.70) = \$500 ÷ 70 | **P* = \$7.14** |
| **6** | Check: value = \$714.29; equity = \$714.29 − \$500 = \$214.29; \$214.29 ÷ \$714.29 | **= 30% exactly** |




Margin percentage as the share price changes, for 100 shares bought with a \$500 loan. The curve falls faster as the price drops because the loan does not shrink.

**Meeting the call.** At \$7 the equity is \$200, but 30% of \$700 is \$210. Depositing **\$10** of cash, which pays down the loan to \$490, restores the margin to exactly 30% (\$210 ÷ \$700). Many brokers ask for more and require the account to be brought back up to the initial 50%, which needs equity of 0.50 × \$700 = \$350, a deposit of **\$150**. If the investor cannot pay, the broker sells shares at the worst possible moment, after the price has already fallen. That is why margin calls are described as *forced selling*, and why leverage that feels manageable in calm markets can wipe out an account in a crash.

## 6. Part C: Short selling

Short selling lets an investor profit when a price falls. There are four steps. First, the investor **borrows shares** through the broker, which locates them from other margin accounts or institutional lenders; the investor pays a borrow fee, typically 0.3% a year for easy-to-borrow stocks and 20% or more for hard-to-borrow ones. Second, the investor **sells** those borrowed shares at the market price; the proceeds are credited to the account but held as collateral. The investor now owes shares, not dollars. Third, the investor **waits**: a price decline is profit and a price increase is loss. Fourth, the investor **covers** by buying the shares back and returning them to the lender, either voluntarily or because a

margin call forces it.

$$\text{Profit} = (\text{Sale price} - \text{Cover price}) \times \text{Shares} - \text{Borrow fees} - \text{Dividends paid}$$

WORKED EXAMPLE 5

### Shorting 100 shares of TSLA at $300 for 60 days

| Step | What we compute | Result |
|---|---|---|
| **1** | Proceeds from the short sale = 100 × $300 (held as collateral) | **$30,000** |
| **2** | Borrow fee at 2% per year for 60 days = $30,000 × 0.02 × 60 ÷ 365 | **$98.63 ≈ $99** |
| **3A** | Scenario A, cover at $240: gross gain = ($300 − $240) × 100 | **+$6,000** |
| | Net profit = $6,000 − $99 | **+$5,901** |
| **3B** | Scenario B, cover at $380: gross loss = ($300 − $380) × 100 | **−$8,000** |
| | Net result = −$8,000 − $99 | **−$8,099** |
| **4** | Break-even cover price: the price fall must pay for the fee, $300 − $99 ÷ 100 | **≈ $299.01** |

The two scenarios involve price moves of similar size ($60 down versus $80 up), and the example also shows the basic **asymmetry of shorting**. The most a short can earn is 100% of the sale price, if the stock goes to zero. There is no ceiling on the loss, because there is no ceiling on how high a stock can rise. In Scenario B the loss keeps growing for as long as TSLA keeps climbing. Short sellers also face risks a buyer does not. In a **short squeeze** (GameStop, January 2021), rising prices force short sellers to cover, and their buying pushes the price higher still. The lender can **recall** the shares at any time. And any **dividend** paid while the position is open must be paid by the short seller to the lender.

## 7. Part C: Inverse ETFs and volatility decay

**SQQQ**, ProShares UltraPro Short QQQ, targets **−3× the daily return** of the NASDAQ-100. It lets an investor bet against the index without borrowing shares, and the most it can lose is what was paid for it. Its long-run expected return is negative, however. The stock market drifts upward over time, and on top of that the daily reset produces **volatility decay**, which is the most important idea in this lecture. SQQQ is a tool for short, well-timed hedges. In 2022 it gained about +79% while QQQ fell 33%, but capturing that required getting both the entry and the exit right.

WORKED EXAMPLE 6

## Two volatile days that go nowhere

Start with $1,000 in QQQ, in a 3× fund, and in a −3× fund. Day 1 the index rises 10%; Day 2 it falls 10%.

| | QQQ (1×) | 3× fund (TQQQ) | −3× fund (SQQQ) |
|---|---|---|---|
| Day 1 return | +10% | 3 × 10% = +30% | −3 × 10% = −30% |
| Value after Day 1 | $1,000 × 1.10 = $1,100 | $1,000 × 1.30 = $1,300 | $1,000 × 0.70 = $700 |
| Day 2 return | −10% | 3 × (−10%) = −30% | −3 × (−10%) = +30% |
| Value after Day 2 | $1,100 × 0.90 = $990 | $1,300 × 0.70 = $910 | $700 × 1.30 = $910 |
| Two-day return | (1.10)(0.90) − 1 = **−1.0%** | (1.30)(0.70) − 1 = **−9.0%** | (0.70)(1.30) − 1 = **−9.0%** |
| Naive '3 × index' | n/a | 3 × (−1%) = −3% | −3 × (−1%) = +3% |

The index lost only 1%, yet the 3× fund lost 9%, three times more than the naive −3%. More striking, the *inverse* fund, which an investor might expect to gain when the index falls, *also* lost 9%. Both leveraged funds lost because a gain followed by an equal percentage loss always leaves you below where you started, and leverage magnifies both legs: $(1 + x)(1 - x) = 1 - x^2$. For the index that is $1 - 0.10^2 = 0.99$; for either ±3× fund it is $1 - 0.30^2 = 0.91$. The loss grows with the *square* of the daily move, which is why decay grows so quickly with leverage and volatility.

WORKED EXAMPLE 7

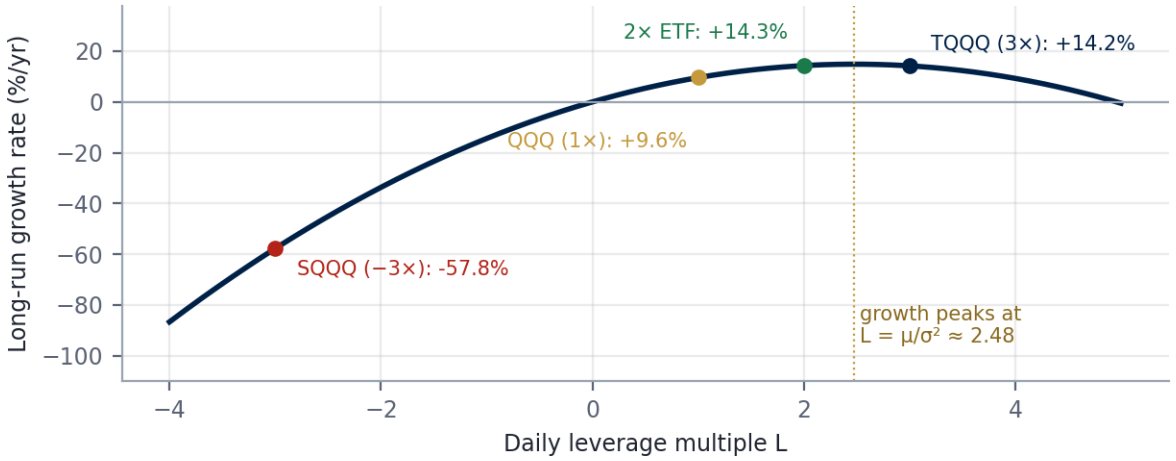
## Volatility drag over a full year (the class simulator)

The simulator in class used 22% annual volatility, a +12% annual drift and 252 trading days. A useful approximation for the long-run growth rate of a fund that resets to leverage L every day, ignoring fees and financing costs, is:

$$g(L) \approx L \times \mu - \tfrac{1}{2} \times L^2 \times \sigma^2$$

The first term is what investors expect: leverage multiplies the drift. The second term is the volatility drag, and it grows with the *square* of leverage. With $\mu = 12\%$ and $\sigma = 22\%$, $\tfrac{1}{2}\sigma^2 = \tfrac{1}{2} \times 0.0484 = 0.0242$, or 2.42% per year for the unleveraged index.

| Fund | L × μ | ½ × L² × σ² | Growth rate g | $1 grows to ≈ $e^g$ |
|---|---|---|---|---|
| QQQ (1×) | 1 × 12% = 12% | 1 × 2.42% = 2.42% | +9.58% | $1.10 |
| 2× ETF | 2 × 12% = 24% | 4 × 2.42% = 9.68% | +14.32% | $1.15 |
| TQQQ (3×) | 3 × 12% = 36% | 9 × 2.42% = 21.78% | +14.22% | $1.15 |
| SQQQ (−3×) | −3 × 12% = −36% | 9 × 2.42% = 21.78% | −57.78% | $0.56 |




Approximate long-run growth rate as a function of daily leverage, with μ = 12% and σ = 22%, before fees and financing costs.

Three lessons follow. First, going from 2× to 3× leverage adds *nothing* to long-run growth under these assumptions: the extra 12 points of drift are eaten by the extra 12.1 points of drag. Growth peaks at L = μ ÷ σ² = 0.12 ÷ 0.0484 ≈ 2.48, and beyond that more leverage lowers growth. Second, adding the 0.88% fee and the financing cost on the borrowed exposure makes 3× look worse still. Third, the inverse fund is hit twice: its drift term is negative in a rising market, *and* it pays the same volatility drag as TQQQ. That is why SQQQ is expected to lose most of its value over a year of normal markets.

**Recovery math: why drawdowns hurt so much**

After a loss of x, you need a gain of x ÷ (1 − x) just to get back to where you started. A 33% loss (QQQ in 2022) needs 0.33 ÷ 0.67 = **+49%**. A 50% loss needs **+100%**. A 79% loss, roughly TQQQ's decline in 2022, needs 0.79 ÷ 0.21 = **+376%**. Leverage makes drawdowns deeper, and the required recovery rises much faster than the drawdown itself.

## 8. Key takeaways

**Leverage is a multiplier in both directions.** Borrowing to invest raises the return on your equity when the asset earns more than the borrowing rate, and deepens the loss when it does not. The ROE formula r + (D/E)(r − i) applies whether the leverage comes from a margin loan or from inside a 3× ETF.

**A daily reset changes everything.** TQQQ is 3× the *daily* return, not 3× the return over time. Because volatility drag grows with the square of leverage, a 3× fund can trail three times the index return even in a rising market, and lose heavily in a volatile, sideways market.

**Margin calls turn investors into forced sellers.** The loan is fixed and the collateral is not, so a price decline eats into equity at a leveraged rate. The margin-call price $P^* = L \div [Q(1 - m)]$ tells you in advance where that forced selling begins.

**Short selling has asymmetric risk.** The gain is capped at 100% and the loss is unlimited, and short squeezes, share recalls and dividends owed add risks that buyers never face.

**Leveraged products matter for the whole system.** The IMF, SEC and FINRA all treat leveraged ETFs as a source of market-wide risk because their end-of-day rebalancing feeds back into prices.

**What a fund holds is not what it is exposed to.** TQQQ holds T-bills and swap contracts, not a single share of Apple or NVIDIA, yet it carries three times the NASDAQ-100's daily risk. Analyzing a portfolio's risk means looking through the holdings to the economic exposure.

# 9. Practice problems

Work these out before reading the solutions. Each one uses a method from the examples above.

**Problem 1.** You buy $20,000 of stock with $8,000 of your own money and borrow the rest at 5%. The stock returns +15% over the year. What is your return on equity? What asset return would leave your ROE at exactly 0%?

**Problem 2.** You buy 200 shares at $50, borrowing $4,000. Maintenance margin is 25%. At what price do you receive a margin call? What is your margin if the price falls to $30?

**Problem 3.** You short 50 shares at $120 and cover 90 days later at $105. The borrow fee is 3% per year and the company paid a $1.00 per share dividend while you were short. What is your net profit?

**Problem 4.** An index rises 5% on Monday and falls 5% on Tuesday. What is the two-day return of the index, of a 2× daily fund, and of a −2× daily fund?

## Solutions

| | Work | Answer |
|---|---|---|
| **P1** | Loan = $20,000 − $8,000 = $12,000, so D/E = 12,000 ÷ 8,000 = 1.5. ROE = 15% + 1.5 × (15% − 5%) = 15% + 15%. Check in dollars: gain $3,000 − interest $600 = $2,400; $2,400 ÷ $8,000 = 30%. | **ROE = 30%** |
| | ROE = 0 requires r + 1.5(r − 5%) = 0, so 2.5r = 7.5% and r = 3%. Check: $600 gain − $600 interest = 0. | **r = 3%** |
| **P2** | P* = L ÷ [Q(1 − m)] = $4,000 ÷ (200 × 0.75) = $4,000 ÷ 150 | **P* = $26.67** |
| | At $30: value = $6,000; equity = $6,000 − $4,000 = $2,000; margin = $2,000 ÷ $6,000 | **33.3%, no call yet** |
| **P3** | Gross gain = ($120 − $105) × 50 | **$750** |
| | Borrow fee = 50 × $120 × 3% × 90 ÷ 365 = $6,000 × 0.03 × 0.2466 | **$44.38** |
| | Dividends owed = 50 × $1.00 | **$50.00** |
| | Net = $750 − $44.38 − $50.00 | **+$655.62** |
| **P4** | Index: (1.05)(0.95) − 1 = 1 − $0.05^2$ − 1 | **−0.25%** |
| | 2× fund: (1.10)(0.90) − 1 = 1 − $0.10^2$ − 1 | **−1.00%** |
| | −2× fund: (0.90)(1.10) − 1 | **−1.00%** |
| | Both leveraged funds lose 4× what the index lost: the drag scales with $L^2 = 4$. | |

Figures on index performance, fund characteristics and regulatory statements are those presented in the Lecture 02 slides. The growth-rate approximation in Example 7 is a standard continuous-time result and ignores fees, financing costs and path details.